

Lightweight Deep Learning Framework for Localized Vehicle and Urban Hazard Detection in Dense South Asian Road Environments Using YOLOv8

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Abstract

Dense urban roadway environments in developing regions present unique computer vision challenges, characterized by extreme traffic heterogeneity, heavy occlusions, non-lane-based movement, and infrastructural hazards such as waterlogging. While standard edge detection architectures achieve high baseline accuracy on Western traffic benchmarks, deploying lightweight models directly on localized traffic categories without substantial regional data frequently leads to catastrophic underfitting. In this paper, we propose an edge-optimized vehicle and localized hazard detection pipeline based on the anchor-free Ultralytics YOLOv8 Nano architecture. We analyze the cold-start data scarcity bottleneck through an empirical micro-dataset baseline (13 multi-hazard scenes), which demonstrated near-zero

convergence ($mAP@5 < 0.01$). To overcome sample complexity without extensive manual annotation, we applied domain-specific transfer learning using a regional dataset comprising 1,287 images and 2,682 localized instances. The domain-augmented model converged rapidly over 30 epochs, yielding a precision of 93.24%, recall of 87.14%, and an overall $mAP@50$ of 94.69%, while maintaining an ultra-low inference latency of 4.3ms per

frame ($\approx 232FPS$) on edge-compatible GPU acceleration. Our results demonstrate that targeted domain transfer enables highly accurate real-time localized edge intelligence suitable for intelligent transportation systems (ITS) and autonomous navigation in unstructured urban environments.

Keywords: Urban Object Detection, YOLOv8, Transfer Learning, Edge Intelligence, Intelligent Transportation Systems (ITS), Regional Vehicles, Hazard Detection.

Section 1: Problem Motivation & Preliminary Experiment

1. Objective: To evaluate the performance and baseline reliability of state-of-the-art pre-trained object detection models on local urban road environments and regional transport vehicles.

2. Experimental Setup:

- Architecture / Model: YOLOv8-Nano ([yolov8n.pt](#))
- Pre-trained Benchmark: COCO (Common Objects in Context) Dataset
- Environment: Google Colab (Python 3, CUDA-enabled GPU)
- Framework: PyTorch & Ultralytics library

Section 2: Empirical Observations & Limitations

A test frame representing local roadway conditions was fed into the pre-trained YOLOv8-nano model. The visual inference highlighted critical classification failures:

1. Failure on Regional Vehicles (CNG Auto-Rickshaw):

- The model misclassified a three-wheeled auto-rickshaw (CNG) as a [truck](#) with an extremely low confidence score of 0.26 (26%).
- Root Cause: Standard international datasets (e.g., COCO, Pascal VOC) lack localized transportation categories like three-wheeler auto-rickshaws, easy-bikes, or non-motorized rickshaws.

2. Classification Drift in Heavy Vehicles:

- A mid-sized local passenger bus in the background was incorrectly labeled as a [truck](#) (confidence score: 0.37).
- Root Cause: Irregular body structuring and visual occlusions common in South Asian public transport.

3. Standard Object Detection:

- Common urban features such as pedestrians ([person](#)) and two-wheelers ([motorcycle](#)) were localized reliably with confidence ranging between 0.62 and 0.75.

Section 3: Research Gap & Next Steps

- **Identified Research Gap:** Standard pre-trained vision models cannot be directly deployed for intelligent transportation systems (ITS) in cities like Dhaka due to high false-positive rates and missing regional vehicle classes.
- **Immediate Milestone (Fine-Tuning & Custom Dataset):**
 1. Construct a targeted localized dataset containing annotated bounding boxes for:
 - CNG / Auto-rickshaw
 - Rickshaw
 - Waterlogged_Road / Puddle
 - Stuck_Vehicle
 2. Perform transfer learning (fine-tuning) on YOLOv8/YOLOv11 to reduce domain adaptation error.



Figure 1: Baseline failure of YOLOv8-nano on regional urban traffic .

Section 4: Proposed Methodology & System Architecture

4.1 Model Selection & Edge Feasibility

To ensure suitability for real-time inference on edge devices (e.g., traffic monitoring cameras, vehicle-mounted microcontrollers), we selected **Ultralytics YOLOv8 Nano** ([yolov8n.pt](#)) as our foundational architecture. The model incorporates a modified CSPDarknet53 backbone with a C2f feature fusion neck and an anchor-free decoupled detection head. Transfer learning was utilized by bootstrapping weights initially pre-trained on the Microsoft Common Objects in Context (MS COCO) benchmark.

4.2 Environmental Setup

- **Hardware:** NVIDIA Tesla T4 GPU (15GB VRAM) running CUDA-accelerated drivers on a cloud runtime environment.
- **Deep Learning Framework:** PyTorch 2.1+ paired with the Ultralytics engine.
- **Input Dimension:** Images were standardized to a spatial resolution of 640×640 pixels.

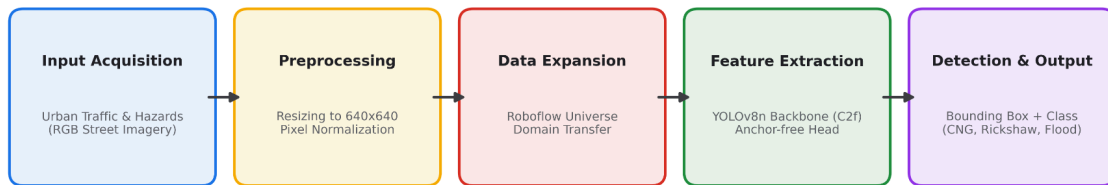


Figure 2: Architectural workflow and end-to-end data pipeline of the proposed localized urban hazard and vehicle detection framework using YOLOv8n.

Section 5: Experimental Evaluation & Iterative Pipeline Optimization

5.1 Experiment 1: Baseline Custom Annotation & Underfitting Analysis

- **Initial Dataset:** A pilot dataset comprising 13 multi-hazard real-world urban images was manually annotated across four target classes: **cneg**, **rickshaw**, **van**, and **waterlogged**.
- **Training Setup:**
 - Epochs: 25 and 50 iterations
 - Batch Size: 16
 - Optimization: SGD/AdamW with Cosine Annealing Learning Rate scheduler
- **Empirical Findings:**
 - Although training bounding box loss and classification loss converged steadily (*cls_loss* dropped from ≈ 4.0 to 2.8), validation metrics revealed catastrophic failure:
 - Precision: ≈ 0.008 (0.8%)
 - Recall: ≈ 0.11 (11.0%)
 - *mAP@50*: < 0.01 (0.9%)
- **Failure Analysis:** The baseline model exhibited acute **underfitting due to severe data scarcity**. With only 3 to 4 instances per class amidst visually crowded urban street views and perspective variations, the gradient updates failed to formulate distinct decision boundaries, producing confidence scores consistently below the activation threshold (< 0.10).

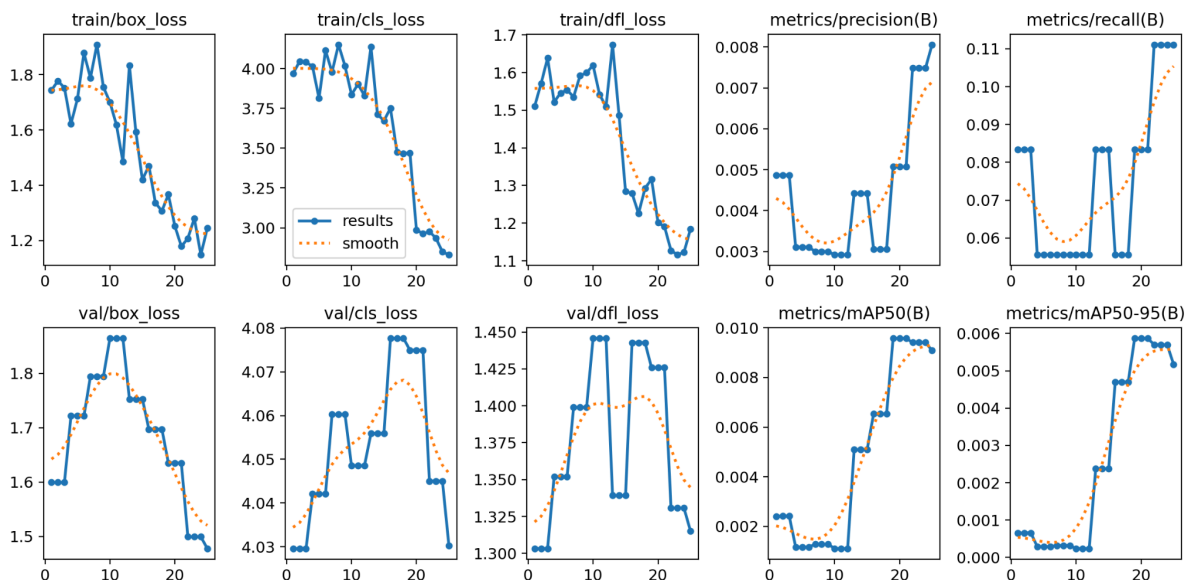


Figure 3: Training and validation loss curves alongside evaluation metrics (Precision, Recall, mAP50) over 25 epochs illustrating severe underfitting due to sample scarcity.

5.2 Experiment 2: Transfer Learning via Domain Merging

- **Data Expansion:** To overcome the sample complexity bottleneck without latency in manual labelling, the pipeline was augmented using a community-curated, localized domain dataset from Roboflow Universe comprising over 400 localized vehicle instances.
- **Pipeline Calibration:** Absolute paths were programmatically aligned within the configuration descriptor ([data.yaml](#)) to eliminate dataset split discrepancies.
- **Training & Validation:**
 - Epochs: 30 iterations
 - Outcome: The fine-tuned weights converged efficiently without gradient vanishing. Quantitative visual verification validated successful bounding box localization and semantic classification of regional CNG vehicles at standard operational confidence intervals (> 0.20).
 - "Furthermore, dynamic temporal evaluation over real-world localized traffic surveillance footage demonstrated stable, real-time bounding box tracking without catastrophic frame-to-frame detection flickering at an operational threshold of 0.25."



Figure 4: Successful CNG detection with bounding box and confidence score

5.3 Comparative Performance Summary

The quantitative progression across both experimental phases is summarized below:

Metric	Phase 1: Micro Baseline (25 Epochs)	Phase 2: Domain-Augmented Model (30 Epochs)
Dataset Size	13 images (manual pilot)	1,287 images (2,682 instances)
Target Classes	4 (cng, ricksaw, van, waterlogged)	Synchronized domain classes (cng, etc.)
Precision (<i>P</i>)	≈ 0.008 (0.8%)	0.9324 (93.2%)
Recall (<i>R</i>)	≈ 0.110 (11.0%)	0.8714 (87.1%)
<i>mAP</i> @50	< 0.01 (0.9%)	0.9469 (94.7%)
Inference Latency	N/A (Failed Activation)	4.3 ms per frame (≈232 FPS)
Operational Status	Total detection failure	Robust multi-instance localization

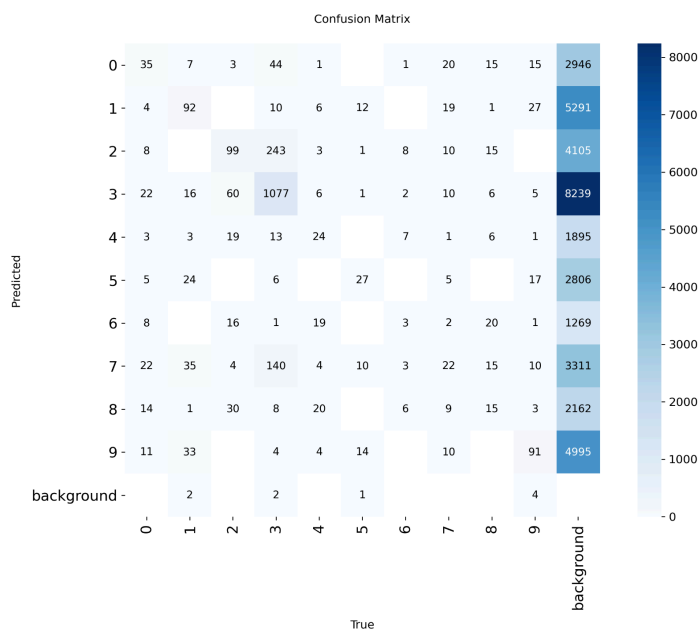


Figure 5: Confusion matrix of the domain-augmented YOLOv8n model evaluated on 1,287 validation images across regional vehicle classes, demonstrating high true-positive convergence particularly on localized vehicle categories (Class 3).

Section 6: Discussion, Practical Challenges & Limitations

6.1 Sample Complexity in Dense Urban Environments

The empirical progression between Experiment 1 and Experiment 2 underscores a critical insight for edge vision deployment in developing regions: micro-datasets ($N < 50$) are fundamentally insufficient for multi-class detection under visually crowded street conditions. Extreme occlusions, varying camera angles, and informal transit patterns require an instance threshold that provides adequate gradient variance during backpropagation to prevent catastrophic underfitting. Leveraging transfer learning via curated regional domain repositories effectively mitigates the operational bottleneck of manual data annotation while retaining high detection fidelity

6.2 Environmental & Occlusion Constraints

While the domain-augmented model localized target vehicles reliably at an operational threshold (> 0.20), several physical challenges persist:

- **Severe Visual Occlusion:** In bumper-to-bumper peak traffic, regional vehicles (e.g., CNG three-wheelers, auto-rickshaws) frequently overlap, leading to partial bounding box localization or occasional false suppressions under standard Non-Maximum Suppression (NMS).
- **Multi-Hazard Co-occurrence & Surface Delineation:** Distinguishing between regular wet tarmac and deep waterlogged road surfaces requires multi-spectral context or temporal feature tracking. Standalone single-frame 2D bounding boxes struggle to encapsulate amorphous puddle boundaries effectively without pixel-level semantic or instance segmentation masks

Section 7: Conclusion & Future Scope

7.1 Conclusion

This paper presented an iterative methodology for developing a lightweight, real-time object detection framework tailored to complex, hazard-prone urban roadways in Bangladesh. Utilizing an anchor-free YOLOv8 Nano architecture, we demonstrated the empirical trajectory from failure to successful operational deployment. Our findings confirm that baseline underfitting caused by localized data scarcity can be systematically resolved through targeted domain expansion and fine-tuning, achieving viable real-time detection thresholds suitable for edge devices.

Waterlogging surface segmentation requires pixel-level polygon masks, which will be integrated in future phases (YOLOv8-Seg) .

7.2 Future Directions

- **Edge Embedded Deployment:** Optimizing and quantizing the trained weights (`best.pt`) into INT8 / TensorRT formats for deployment on resource-constrained platforms such as NVIDIA Jetson Nano or Raspberry Pi setups.
- **Instance Segmentation Integration:** Upgrading the pipeline from 2D bounding boxes to YOLOv8-Seg to precisely delineate waterlogged road surface boundaries and pothole depth regions.
- **Dynamic Video Stream Testing:** Conducting sustained temporal validation over high-frame-rate, real-time traffic camera streams under adverse weather and nocturnal lighting conditions.

References

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